Fiscal Year:	FY 2020	**Task Last Updated:**	FY 05/10/2021
PI Name:	Mohraz, Ali Ph.D.		
Project Title:	New Bijel Chemistries for Long-duration Microgravity and Ground-based Experiments		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Irvine	**State:**	CA
Zip Code:	92697-2575	**Congressional District:**	45
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2013 Complex Fluids & Macromolecular Biophysics NNH13ZTT001N
Start Date:	08/06/2018	**End Date:**	01/05/2021
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	Chao, David	**Contact Phone:**	216-433-8320
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC18K1554		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	This is ongoing university science support for Advanced Colloids Experiments (ACE) colloids microgravity research experiment on the International Space Station (ISS). The previous grant, "Understanding the Morphology and Stability of Bijels Using Microgravity," was grant number NNX13AQ69G and was flight-approved. This continuation flight project is scheduled for flight in February 2019. This new grant will accomplish the scheduled 02/2019 launch and operating the science experiment on International Space Station. The overarching aim of the proposed program is to gain a better fundamental understanding of the relationship between the microstructure, rheology, stability, and process ability of bijels through synergistic, long-duration ground-based and microgravity experiments and theoretical calculations, using new bijel chemistries that are safe for microgravity research and nonvolatile, allowing long-duration rheology experiments.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Bicontinuous interfacially jammed emulsion gels (bijels) have a unique microstructure that is suitable for the development of advanced materials and composites that require many sample-spanning phases to co-exist. For example, next generation battery electrodes are envisioned as 3D arrangements of ion- and electron-conducting phases in coexistence with electrolyte channels that percolate throughout the entire electrode volume. This morphology can be uniquely realized in bijels, as they inherently enjoy a co-continuous microstructure that is self-assembled through an external stimulus (temperature change). Similarly, the co-continuous morphology of bijel-derived materials can prove beneficial for tissue engineering and cell delivery scaffolds or any other structure that requires microbial growth, such as electrodes in microbial fuel cells. Finally, the open pore morphology and uniform domain size uniquely realized in bijels can lead to better hydraulic permeability for a given porosity, leading to enhanced transport and lower energy consumption for fluid flow. Therefore, bijel processing will provide a unique route to advanced energy and health care materials, and understanding the rheology and processability of bijels, which is the focus of this project, will pave the way for the realization of this end goal. Applications of bijels in each of these areas are currently being investigated in our laboratory.
Task Progress:	NOTE (Ed., May 2021): Compiled from Progress report submitted late 2019; reporting covers through that time period: The overarching objective of the research program is to assess the mechanical stability, rheology, and processability of solid-stabilized emulsions such as high internal phase emulsions (HIPEs), Pickering emulsions, and bijels (bicontinuous interfacially jammed emulsion gels), and investigate the relationship between their microstructure and rheology, while also investigating and expanding the Earth benefits of bijel-derived materials. In addition, new bijel chemistries are sought to enable safe handling and experiments with bijels at the International Space Station (ISS), and long-duration rheology experiments on Earth. This will be accomplished through a combination of ground and microgravity experiments, and basic modeling of solid-stabilized emulsion rheology. The major activities during year one of the project have involved strategies to tune the evolution of particle surface chemistiy during aging to allow bijel formation at the ISS, investigation of the microstructure of bijel-derived porous materials, development of bijel-derived biomaterials for tissue integration, and fundamental studies of the stability of solid-stabilized high internal phase emulsions (HIPEs). The findings from areas Microstructure of Bijel-Derived Materials, and Bijel-Derived Biomaterials for Enhanced Tissue Integration) form the bases of two peer-reviewed manuscripts that were accepted for publication during the review period, in Acta Biomaterialia [published August 2019--see Bibliography below (Ed., 5/10/21)], and Materialia, respectively. We are also in the final stages of editing two other manuscripts based on work that has been supported by this award, with target submission dates within the next month, and in the early stages of preparing another manuscript on instant bijel scaffolds, with a target submission date within the next three months.
Bibliography Type:	Description: (Last Updated: 04/22/2025)
Articles in Peer-reviewed Journals	Thorson TJ, Gurlin RE, Botvinick EL, Mohraz A. "Bijel-templated implantable biomaterials for enhancing tissue integration and vascularization." Acta Biomater. 2019 Aug;94:173-82. https://doi.org/10.1016/j.actbio.2019.06.031 ; PMID: 31233892; PMCID: PMC7433011 , Aug-2019